

Worksheet 4

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Branch: MCA

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Subject Name: Internet of Things

UID: 25MCA20078

Section/Group: 25MCA-1(A)

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1. Aim/Overview of the practical:

Interface an Arduino Uno, Led and Buzzer with RFID reader/writer MFRC522 and register the card's UID.

Write a program to check whether the card used is authorized or unauthorized.

- If the card is authorized then Turn ON only "Green Led" and buzzer will Beep Once.
- If the card is unauthorized then Turn ON only "Red Led" and buzzer will Beep three times.

2. Apparatus (For applied/experimental sciences/materials based labs):

Hardware Requirements

- Arduino Uno
- Ultrasonic Sensor (HC-SR04)
- 16×2 LCD
- Green LED
- Yellow LED
- Buzzer
- Resistors (220Ω for LEDs)
- Breadboard and jumper wires

Software requirements

- Arduino IDE
- Tinkercad Circuits

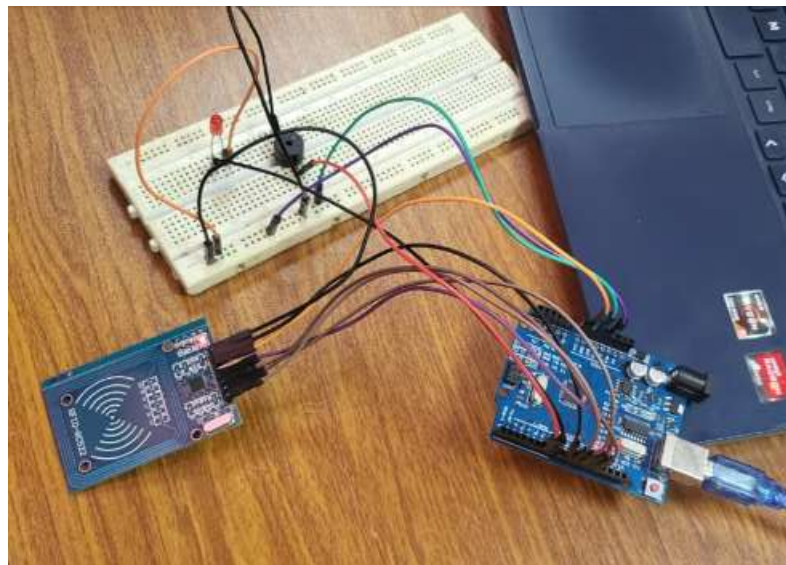
3. Circuit Diagram (TinkerCad):

Code:

```
#include <SPI.h>
#include <MFRC522.h>
#define SS_PIN 10
```

```
#define RST_PIN 9
#define LED_PIN 7
#define BUZZER_PIN 6
MFRC522 rfid(SS_PIN, RST_PIN);
// CHANGE THIS TO YOUR CARD UID
byte authorizedUID[] = {0xF9, 0x3A, 0xCA, 0x79};
void setup() {
    Serial.begin(9600);
    SPI.begin();
    rfid.PCD_Init();
    pinMode(LED_PIN, OUTPUT);
    pinMode(BUZZER_PIN, OUTPUT);
    digitalWrite(LED_PIN, LOW);
    Serial.println("RFID System Ready");
    Serial.println("Scan your card...");
}
void loop() {
    // Check for a new card
    if (!rfid.PICC_IsNewCardPresent()) {
        return;
    }
    // Read card
    if (!rfid.PICC_ReadCardSerial()) {
        return;
    }
    Serial.print("Card UID: ");
    for (byte i = 0; i < rfid.uid.size; i++) {
        Serial.print(rfid.uid.uidByte[i], HEX);
        Serial.print(" ");
    }
    Serial.println();
    // Check whether card is authorized
    bool authorized = true;
    if (rfid.uid.size != sizeof(authorizedUID)) {
        authorized = false;
    }
    else {
        for (byte i = 0; i < rfid.uid.size; i++) {
            if (rfid.uid.uidByte[i] != authorizedUID[i]) {
                authorized = false;
                break;
            }
        }
    }
    // Authorized card
    if (authorized) {
```

```
Serial.println("ACCESS GRANTED");  
digitalWrite(LED_PIN, HIGH);  
tone(BUZZER_PIN, 1000);  
delay(300);  
noTone(BUZZER_PIN);  
delay(2000);  
digitalWrite(LED_PIN, LOW);  
}  
// Unauthorized card  
else {  
    Serial.println("ACCESS DENIED");  
    for (int i = 0; i < 3; i++) {  
        digitalWrite(LED_PIN, HIGH);  
        tone(BUZZER_PIN, 300);  
        delay(200);  
        digitalWrite(LED_PIN, LOW);  
        noTone(BUZZER_PIN);  
        delay(200);  
    }  
}  
rfid.PICC_HaltA();  
rfid.PCD_StopCrypto1();  
delay(1000);  
}
```



```
ACCESS GRANTED  
Card UID: F9 3A CA 79  
ACCESS GRANTED  
Card UID: F9 3A CA 79  
ACCESS GRANTED  
Card UID: F9 3A CA 79  
ACCESS GRANTED  
RFID System Ready  
Scan your card...  
RFID System Ready  
Scan your card...
```

```
Output Serial Monitor X  
Message (Enter to send message to 'Arduino UNO' on 'C  
RFID System Ready  
Scan your card...  
RFID System Ready  
Scan your card...  
Card UID: F9 3A CA 79  
ACCESS DENIED  
Card UID: F9 3A CA 79  
ACCESS DENIED  
Card UID: F9 3A CA 79  
ACCESS DENIED  
Card UID: F9 3A CA 79
```

5. Learning outcomes (What I have learnt):

1. Understand RFID card detection.
2. Interface RFID sensor with Arduino.
3. Control LED and buzzer using Arduino.
4. Implement authorized card access using RFID UID.